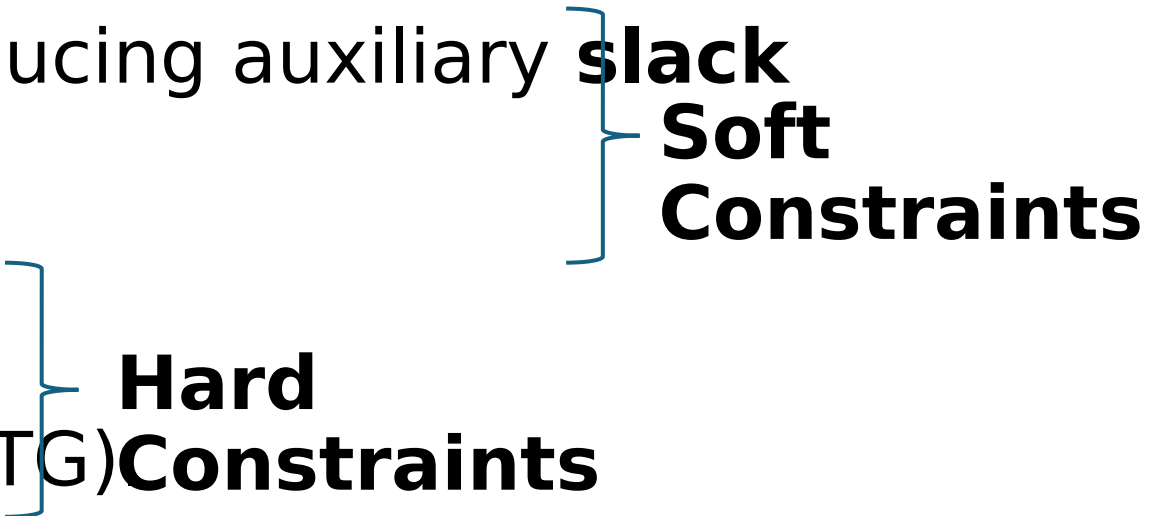


Quantum Computing Approaches for the Quadratic Knapsack Problem (QKP)

Andres Valencia

Background and Related Work

- QUBO formulation by introducing auxiliary **slack variables**.
 - Slack-free QUBO.
 - Biased state preparation.
 - Quantum Tree Generator (QTG)
- Soft Constraints**
- Hard Constraints**
- 
- A diagram illustrating the classification of constraints. A blue bracket on the right groups the first two items of the list ('QUBO formulation by introducing auxiliary slack variables' and 'Slack-free QUBO') under the label 'Soft Constraints'. Another blue bracket on the right groups the last two items ('Biased state preparation' and 'Quantum Tree Generator (QTG)') under the label 'Hard Constraints'.

Background and Related Work

Soft Constraints:

Increases the objective function by penalizing infeasible states (those that do not meet the constraints)

Hard Constraints:

Restricts the class of parameterized states to the feasible subspace (the space spanned by all feasible solutions)

QUBO formulation by introducing auxiliary slack variables

- The idea is to convert the inequality constraint $\sum w_i x_i \leq C$ into an equality constraint using a slack variable s , such that:

$$\sum_{i=1}^n w_i x_i + s = C, \quad \text{with } s = \sum_{k=0}^{m-1} 2^k s_k, \quad s_k \in \{0, 1\}$$

- However, this definition is not unique, there are at least 6 formulation to introduce slack variables.

Translating Constraints into QUBOs for the Quadratic Knapsack Problem

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QUBO formulation by introducing auxiliary slack variables

- We can see that 6 methods require more than n qubits to transform the problem into QUBO. This increases the complexity of the quantum optimization algorithm. Furthermore, they also distract the algorithm from finding optimal solutions because of the penalty terms.
- The selection of the penalty value is special for each instance.

Slack-Free QUBO

- The second category seeks to avoid auxiliary variables entirely by applying penalty-based methods directly on the constraint.
- These slack-free QUBO formulations embed the constraint into the objective function via a penalty term. A general form of the penalized objective is:

$$\max_{x \in \{0,1\}^n} \left[x^T Q x + \lambda \cdot \text{Penalty} \left(\sum_{i=1}^n w_i x_i - C \right) \right]$$

Slack-Free QUBO

- Some ways have been proposed to introduce the penalty into the Hamiltonian. One is to simply design a step function:

$$\text{Penalty}(y) = \begin{cases} 0, & \text{if } y \geq 0 \\ 1, & \text{otherwise} \end{cases}, \text{ where } y = \left(\sum_{i=1}^n w_i x_i - C \right)$$

- But a simple implementation of this requires knowing all the states that satisfy the constraint. And they compute it classically.

Implementing Slack-Free Custom Penalty Function
for QUBO on Gate-Based Quantum Computers

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Slack-Free QUBO

- Other proposals suggest using an exponential function

$$\text{Penalty}(y) = e^{-\lambda y}, \text{ where } y = \left(\sum_{i=1}^n w_i x_i - C \right)$$

- But this type of functions has no implementation in quantum computing, so it is expanded to second order by adding these terms to the Hamiltonian.

Unbalanced penalization: A new approach to encode inequality constraints of combinatorial problems for quantum optimization algorithms

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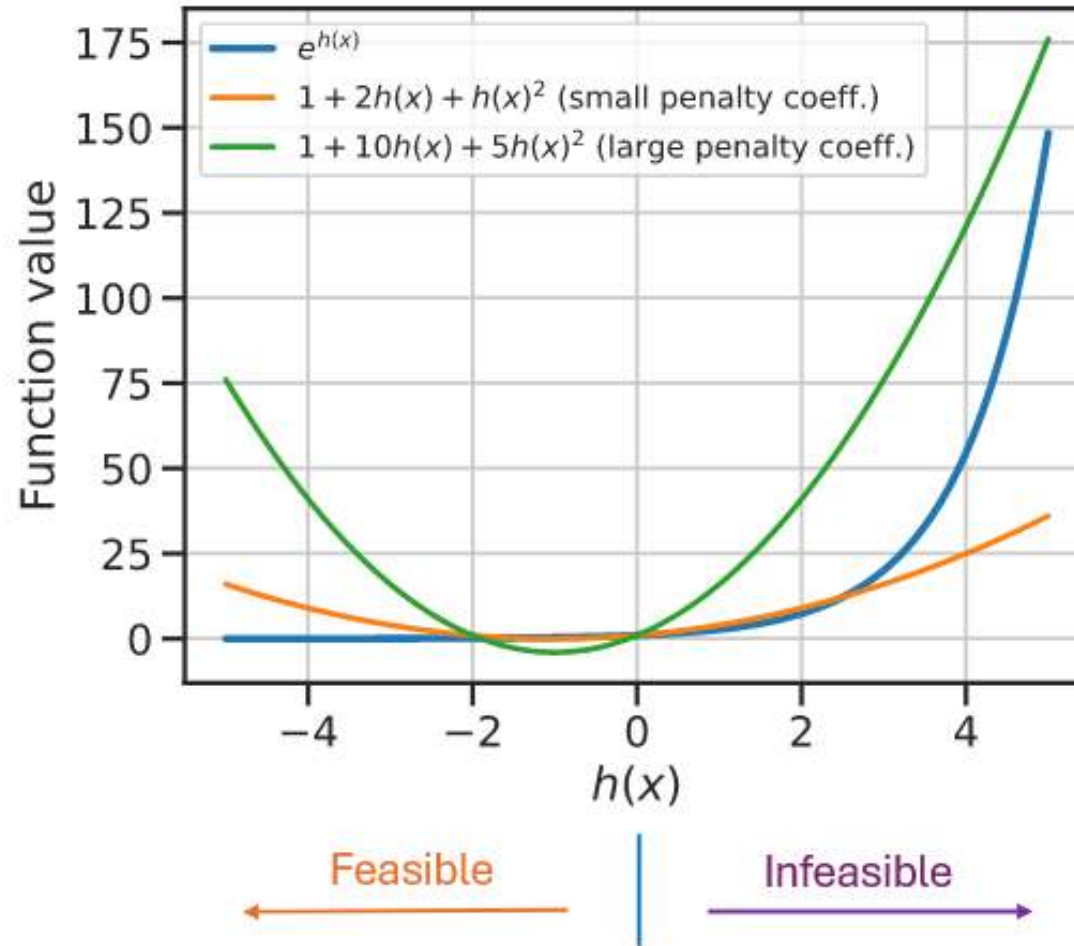
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Slack-Free QUBO

- And the choice of penalty value is a problem also.



Biased state preparation

- This alternative aims to initiate a state that assigns zero or near-zero probabilities to those who do not satisfy the constraint. To achieve this, the amplitudes are given by an exponential form of this type.

superposition of the optimal value. First, we consider the penalty function of the constrained problem as follows:

$$g(x) = f(x) - \chi_F(x) \quad (11)$$

with $\chi_F(x)$ is the indicator function:

$$\chi_F(x) = \begin{cases} 0, & \text{if } x \in F \\ \infty, & \text{otherwise} \end{cases}. \quad (12)$$

If $x \notin F$, $g(x)$ goes to $-\infty$. The aim of our method and penalty function for QAOA have the same role is to define which solution is not in the feasible space. Next, we construct the diagonal matrix H that represents the value of

Biased state preparation

- This has two problems. A QAOA scheme does not guarantee search only within the feasible subspace.
- Creating this initial state requires knowing (classically precomputing) all the states that do not satisfy the constraint.



**Solving Quadratic Knapsack Problem
with Biased Quantum State Optimization
Algorithm**

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**An Angle-expressed Quantum Evolutionary Algorithm for
Quadratic Knapsack Problem**

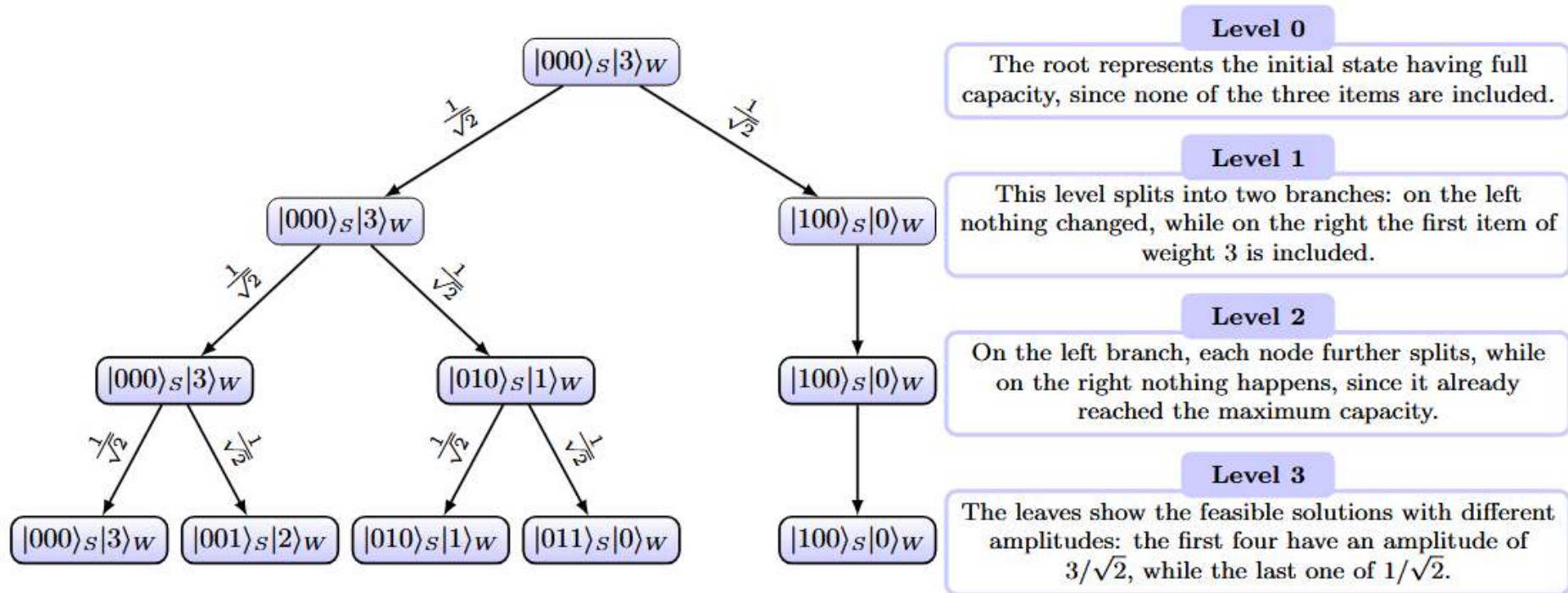
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Quantum Tree Generator (QTG)

The most constraint-aware approach is based on the Quantum Tree Generator. This method generates a superposition exclusively over feasible solutions using a tree-structured circuit where each branching decision respects the weight constraint.



Quantum Tree Generator (QTG)

- This algorithm consists of preparing a feasible state, and then performs a Grover-type search to find the maximum.
- The main paper was for the linear knapsack problem (NIPS 2024)

A quantum algorithm for solving 0-1 Knapsack problems

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Here we present two novel contributions for achieving quantum advantage in solving difficult optimisation problems, both in theory and foreseeable practice. (1) We introduce the “Quantum Tree Generator”, an approach to generate in superposition all feasible solutions of a given instance, yielding together with amplitude amplification the optimal solutions for 0-1 knapsack problems. The QTG offers massive memory savings and enables competitive runtimes compared to the classical state-of-the-art knapsack solvers (such as COMBO, Gurobi, CP-SAT, Greedy) already for instances involving as few as 100 variables. (2) By introducing a new runtime calculation technique that exploits logging data from the classical solver COMBO, we can predict the runtime of our method way beyond the range of existing quantum platforms and simulators, for various benchmark instances with up to 600 variables. Combining both of these innovations, we demonstrate the QTG’s potential *practical* quantum advantage for large-scale problems, indicating an effective approach for combinatorial optimisation problems.

Quantum Tree Generator (QTG)

- The same authors then extended their method to the quadratic and multi knapsack problems (March 2025).
- The main disadvantages of this method are the large number of qubits it uses, since it requires 3 registers, one for the items, another to carry the capacity of the knapsack, and a last one for the gain.
- And they need to define a threshold parameter to fine-tune the search for the maximum.

A quantum search method for quadratic and
multidimensional knapsack problems

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Quantum Tree Generator (QTG)

- Then (Nov 2024) one of the authors along with others, made use of the QTG feasible state preparation to apply a QAOA scheme, in the linear knapsack problem.
- Here, the qubit register for the gain is dispensed with. A mixer is introduced for QAOA that preserves the search within the feasible space.

Quantum tree generator improves QAOA state-of-the-art for the knapsack problem

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(Dated: November 4, 2024)

Quantum Tree Generator (QTG) for QAOA

- In this paper they perform all the results under classical simulations, and facilitate calculations by saving classical information and doing pre-calculation.
- And they only apply it to the linear knapsack case.

tion. Lastly, the phase separator is implemented as for the other QAOA version, that is by pre-computing all objective values and performing an element-wise vector multiplication. Note, however, that in this second ver-

There exist several suitable packages for simulating QAOA instances, e.g. [31]. They are typically based on a state-vector-based simulation: The current state of an n -qubit register is represented by a 2^n -dimensional complex vector and the application of quantum gates is simulated by matrix-vector multiplication, usually in a suitable sparse format (such as in qHiPSTER [32]). We

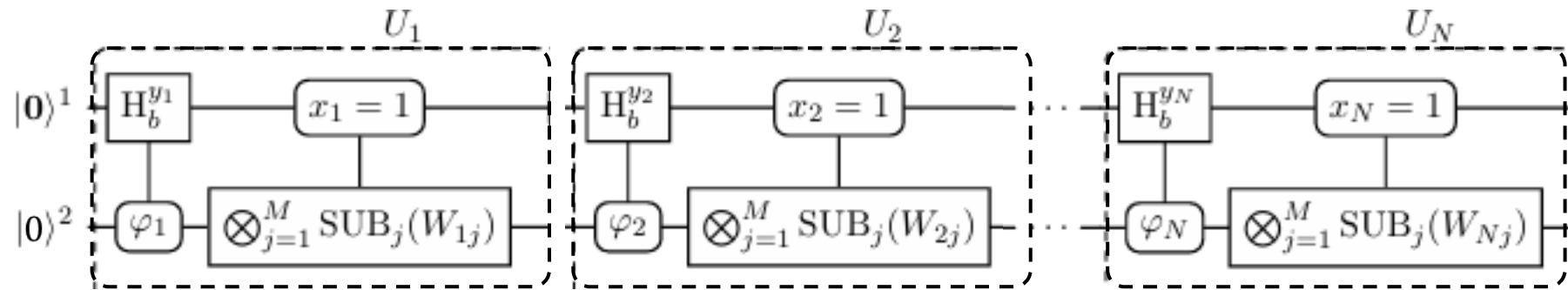
In our case, we additionally have to test for feasibility, since we wish to assign the value zero to infeasible states. Therefore, each bit string is (again classically) tested for feasibility before calculating its objective value.

the simulation, we employ another common trick: We pre-compute the objective values of all 2^n bit strings and store them in an objective vector. When simulating the application of $e^{-i\gamma H_f}$ we perform a element wise prod

Quantum Tree Generator (QTG) for QAOA

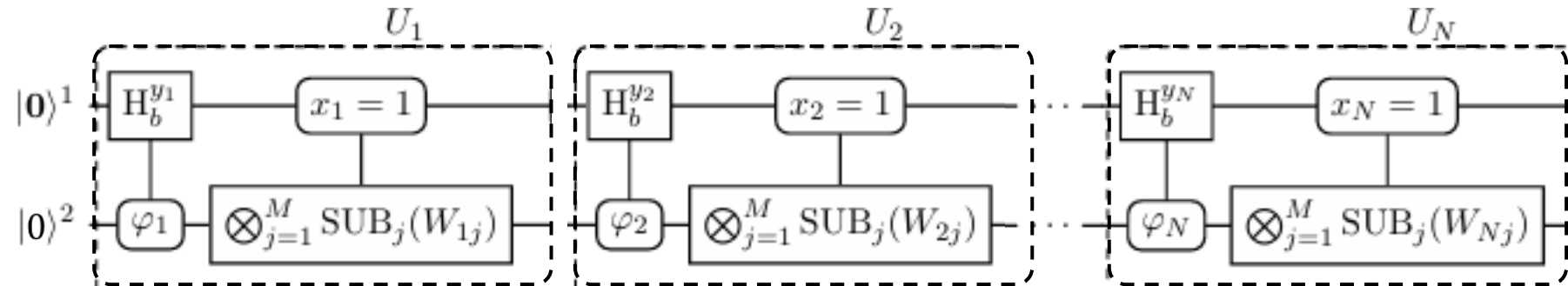
- My proposal is to apply this scheme to the quadratic knapsack case (It has not been done before).
- Implement the simulation in QISKIT. This makes it easier to define the entire classic assembly, and we use the QISKIT simulators directly.
- Do not pre-calculate anything, and do not save classical information.

Quantum Tree Generator (QTG) for QAOA

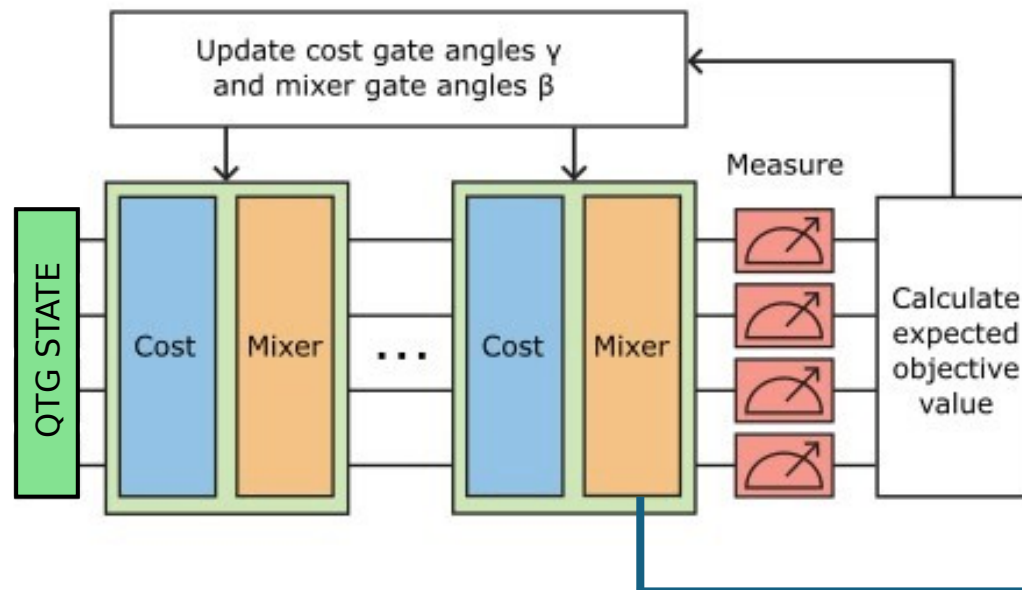


- for each item $i = 1, \dots, N$ do:
 1. Apply the biased Hadamard gate H_{y_i} to $|x_i\rangle^1$ if $w_i \leq C$;
 2. Controlled on $|x_i\rangle^1$, update $|C\rangle^2$ such that $C' = C + w_i$;
- Greedy solution as the intermediate solution to bias the QTG. **Is there a better alternative to QKP?**

Quantum Tree Generator (QTG) for QAOA



The QTG structure is used to perform QAOA optimization



This further allows simulating the action of the QTG-based mixer thanks to the following identity:

$$\begin{aligned}
 U_M(\beta) |\psi\rangle &= (\mathbb{1} - (1 - e^{-i\beta}) |KP\rangle\langle KP|) |\psi\rangle \\
 &= |\psi\rangle - (1 - e^{-i\beta}) \langle KP|\psi\rangle |KP\rangle.
 \end{aligned} \tag{19}$$

Quantum Tree Generator (QTG) for QAOA

To do:

- Research and improve the mixer. Since it is the one that consumes the most resources.
- Investigate the best initial parameters.
- Get results (Requires more computing).
- Write the review and the proposed algorithm.